GATE ME 90 Days Challenge

Day 41/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Heat Transfer	**1-D Conduction**	**64 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on modes of heat transfer, Fourier's law, 1-D steady conduction in plane walls, cylinders and spheres, the thermal-resistance network (series and parallel), variable conductivity, heat generation, contact resistance and the critical radius of insulation. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

In a pure metal such as copper, heat is conducted mainly by

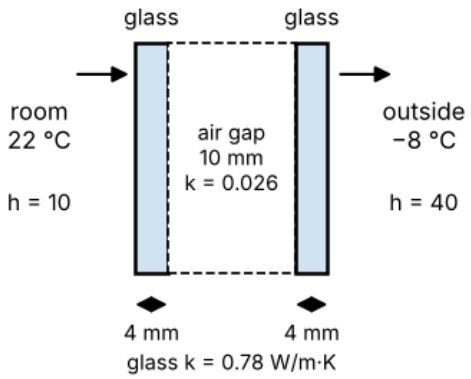
(A) lattice vibrations (phonons) alone
(B) free electrons
(C) radiation between atoms
(D) bulk motion of the atoms

Q2 NAT

A plane wall 0.15 m thick (k = 1.2 W/m·K) has its faces at 150 °C and 30 °C. The steady heat flux (in W/m²) is _____

Q3 NAT

The double-glazed window shown (area 1.5 m²) has two 4 mm glass panes (k = 0.78 W/m·K) separated by a 10 mm stagnant air gap (k = 0.026 W/m·K). The room is at 22 °C (h = 10 W/m²·K) and outside air at −8 °C (h = 40 W/m²·K). The heat loss through the window (in W, round off to one decimal place) is _____



Q4 MCQ

The conduction resistance of a hollow cylinder of inner radius r_1, outer radius r_2, length L and conductivity k is

(A) $(r_2 - r_1)/(kL)$
(B) $\ln(r_2/r_1)/(2\pi kL)$
(C) $(r_2 - r_1)/(4\pi k\, r_1 r_2)$
(D) $2\pi kL/\ln(r_2/r_1)$

Q5 NAT

A tube of inner radius 25 mm and outer radius 35 mm is 2 m long (k = 15 W/m·K). Its inner and outer surfaces are at 120 °C and 100 °C. The heat flow through the tube wall (in W, round off to the nearest integer) is _____

Q6 MCQ

An electric wire whose radius is smaller than the critical radius of insulation is covered with a thin layer of insulation. The effect is that

(A) the heat dissipated decreases and the wire runs hotter
(B) the heat dissipated is unchanged
(C) the heat dissipated increases, so the wire runs cooler for the same current
(D) the wire cannot reach steady state

Q7 NAT

A wire of 3 mm diameter is to be insulated with a material of k = 0.18 W/m·K; the outside heat transfer coefficient is 12 W/m²·K. The insulation thickness (in mm, round off to one decimal place) for maximum heat dissipation is _____

Q8 MCQ

A nuclear fuel rod of radius 10 mm (k = 25 W/m·K) generates heat uniformly at 5×10^7 W/m³. Its surface is held at 300 °C. The centre-line temperature is

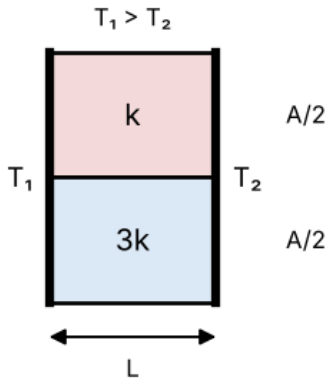
(A) 325 °C
(B) 400 °C
(C) 350 °C
(D) 312.5 °C

Q9 NAT

A flat plate surface at 400 K (emissivity 0.8) loses heat to still air at 300 K (h = 10 W/m²·K) and to large surroundings also at 300 K. Taking $\sigma = 5.67 \times 10^{-8}$ W/m²·K⁴, the total heat loss per m² (in W/m², round off to the nearest integer) is _____

Q10 MCQ

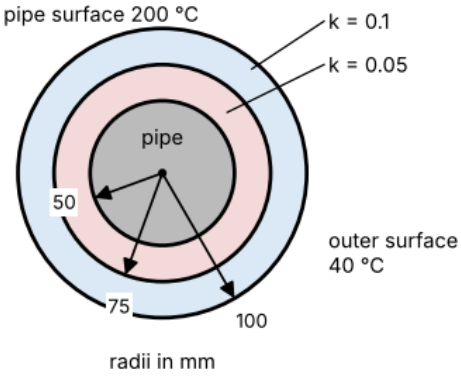
Two slabs of the same thickness L and equal face areas are placed side by side between the same two isothermal surfaces, as shown. Their conductivities are k and 3k. The equivalent conductivity of the combination is



(A) 1.5k
(B) 4k
(C) 0.75k
(D) 2k

Q11 NAT

A steam pipe of outer radius 50 mm is at 200 °C. It carries two insulation layers: the inner one (k = 0.05 W/m·K) up to a radius of 75 mm and the outer one (k = 0.1 W/m·K) up to 100 mm, as shown. The outer surface is at 40 °C. The heat loss per metre (in W/m, round off to one decimal place) is _____



Q12 MCQ

In steady 1-D conduction through a composite plane wall with no heat generation, which quantity is the same for every layer?

(A) Temperature drop
(B) Temperature gradient
(C) Heat flux
(D) Thermal resistance

Q13 NAT

A plane wall 50 mm thick (k = 25 W/m·K) generates heat uniformly at 10^6 W/m³. One face is perfectly insulated and the other is held at 60 °C. The maximum temperature in the wall (in °C) is _____

Q14 MCQ

Thermal contact resistance between two pressed metal surfaces can be reduced by

(A) polishing one surface and roughening the other

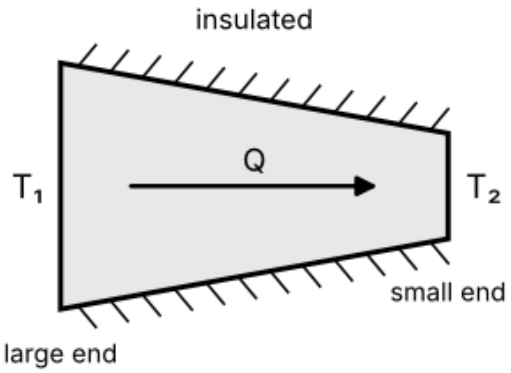
(B) increasing the contact pressure and filling the gaps with a thermal grease

(C) evacuating the gaps between the surfaces

(D) coating the surfaces with a thin plastic film

Q15 MCQ

Heat is conducted steadily along a solid truncated cone with its curved surface insulated, from the larger end to the smaller end, as shown. With constant k, the magnitude of the temperature gradient along the axis is



(A) the same everywhere

(B) largest at the smaller end

(C) largest at the larger end

(D) zero at the smaller end

Answer Key

Q	Type	Answer
Q1	MCQ	(B) free electrons
Q2	NAT	960
Q3	NAT	86.6 (accept 85.8 to 87.4)
Q4	MCQ	(B) $\ln(r_2/r_1)/(2\pi kL)$
Q5	NAT	11204 (accept 11148 to 11260)
Q6	MCQ	(C) the heat dissipated increases, so the wire runs cooler for the same current
Q7	NAT	13.5
Q8	MCQ	(C) 350 °C
Q9	NAT	1794 (accept 1785 to 1803)
Q10	MCQ	(D) 2k
Q11	NAT	91.5 (accept 90.6 to 92.4)
Q12	MCQ	(C) Heat flux
Q13	NAT	110
Q14	MCQ	(B) increasing the contact pressure and filling the gaps with a thermal grease
Q15	MCQ	(B) largest at the smaller end

Detailed Solutions

Q1 **Answer: (B) free electrons**

Concept: Conduction in solids has two carriers: free electrons and lattice vibrations (phonons).

In pure metals the free-electron contribution dominates, which is why good electrical conductors are also good thermal conductors (Wiedemann–Franz law).

Alloying scatters electrons, so alloys (e.g. stainless steel) have much lower k than pure metals.

Exam tip: Order of k: pure metals > alloys > non-metallic solids > liquids > gases.

Q2 **Answer: 960**

Concept: Fourier's law for 1-D steady conduction with constant k: $q = k(T_1 - T_2)/L$.

$q = 1.2 \times (150 - 30)/0.15 = 1.2 \times 800$

$q =$ **960 W/m²**

Exam tip: In a plane wall with constant k and no generation, the profile is a straight line.

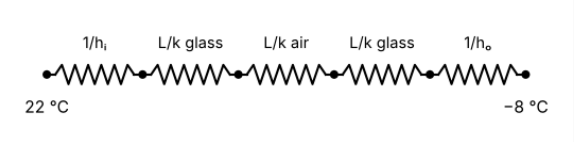
Q3 **Answer: 86.6 (accept 85.8 to 87.4)**

Concept: Series resistances per m² add: $1/h_i + L_g/k_g + L_a/k_a + L_g/k_g + 1/h_o$ (circuit below).

$R'' = 1/10 + 0.004/0.78 + 0.010/0.026 + 0.004/0.78 + 1/40 = 0.1 + 0.00513 + 0.38462 + 0.00513 + 0.025 = 0.5199$ m²K/W

$q = 30/0.5199 = 57.71$ W/m²; $Q = 57.71 \times 1.5 =$ **86.6 W**

Exam tip: The thin air gap gives about 74 % of the total resistance; the glass itself hardly matters.



Q4 **Answer: (B) $\ln(r_2/r_1)/(2\pi kL)$**

Concept: In a cylinder the area $2\pi rL$ grows with r, so $Q = -k(2\pi rL)\,dT/dr$ is integrated with Q constant.

$Q = 2\pi kL(T_1 - T_2)/\ln(r_2/r_1)$, so $R = \ln(r_2/r_1)/(2\pi kL)$

Option (C) is the resistance of a spherical shell; option (D) is the conductance, not the resistance.

Exam tip: For a thin tube ($r_2 \approx r_1$) the cylinder result reduces to the plane-wall value t/(kA).

Q5 **Answer: 11204 (accept 11148 to 11260)**

Concept: $Q = 2\pi kL(T_1 - T_2)/\ln(r_2/r_1)$.

$\ln(35/25) = \ln 1.4 = 0.33647$

$Q = 2\pi \times 15 \times 2 \times 20/0.33647 = 3769.9/0.33647 =$ **11204 W**

Exam tip: Keep radii as a ratio: units cancel inside the logarithm.

Q6 **Answer: (C) the heat dissipated increases, so the wire runs cooler for the same current**

Concept: Insulation adds conduction resistance but also increases the outer surface area, reducing convection resistance $1/(2\pi rLh)$.

For $r < r_c = k/h$ the drop in convection resistance is larger, so the total resistance falls and heat loss rises.

Heat loss is maximum at $r = r_c$; beyond it insulation reduces heat loss.

Exam tip: This is why thin wires are insulated: the insulation helps cooling while giving electrical safety.

Q7 **Answer: 13.5**

Concept: Critical radius for a cylinder $r_c = k/h$.

$r_c = 0.18/12 = 0.015$ m = 15 mm

Thickness $= r_c - r_{wire} = 15 - 1.5 =$ **13.5 mm**

Exam tip: The question asks for thickness, not radius: subtract the wire radius.

Q8 **Answer: (C) 350 °C**

Concept: Solid cylinder with uniform generation: $T(r) = T_s + q_g(R^2 - r^2)/(4k)$, maximum on the axis.

$T_{max} - T_s = q_g R^2/(4k) = 5 \times 10^7 \times 10^{-4}/(4 \times 25) = 50$ °C

$T_{max} =$ **350 °C**

Exam tip: Denominator: 2k for a plane wall (half-thickness L), 4k for a cylinder, 6k for a sphere.

Q9 **Answer: 1794 (accept 1785 to 1803)**

Concept: Convection and radiation act in parallel from the same surface: $q = h(T_s - T_\infty) + \varepsilon\sigma(T_s^4 - T_{sur}^4)$.

Convection: $10 \times 100 = 1000$ W/m²

Radiation: $0.8 \times 5.67 \times 10^{-8} \times (400^4 - 300^4) = 0.8 \times 5.67 \times 10^{-8} \times 1.75 \times 10^{10} = 793.8$ W/m²

Total = **1794 W/m²**

Exam tip: Even at modest temperatures radiation is comparable to free convection; never ignore it unless told.

Q10 **Answer: (D) 2k**

Concept: Side-by-side layers see the same ΔT, so their conductances add (parallel resistances).

$Q = [k(A/2) + 3k(A/2)]\Delta T/L = 2kA\Delta T/L$, so $k_{eq} =$ **2k**

If the same slabs were in series (one after the other, each L/2 thick): $k_{eq} = L/(L/2k + L/6k) = 1.5k$.

Exam tip: Parallel → area-weighted mean of k; series → thickness-weighted harmonic mean.

Q11 **Answer: 91.5 (accept 90.6 to 92.4)**

Concept: Cylindrical layers in series: $R' = \Sigma \ln(r_{out}/r_{in})/(2\pi k)$.

Inner: $\ln(75/50)/(2\pi \times 0.05) = 0.4055/0.3142 = 1.2906$ K·m/W

Outer: $\ln(100/75)/(2\pi \times 0.1) = 0.2877/0.6283 = 0.4579$ K·m/W

$Q' = (200 - 40)/1.7485 =$ **91.5 W/m**

Exam tip: The interface is at $200 - 91.5 \times 1.2906 = 81.9$ °C; check the inner layer is rated for 200 °C.

Q12 **Answer: (C) Heat flux**

Concept: In steady state no energy is stored in any layer, so whatever enters a layer must leave it.

Hence the heat flux q is the same through all layers (series circuit: same 'current').

Temperature drop and gradient differ from layer to layer ($\Delta T = qL/k$).

Exam tip: The layer with the largest L/k takes the largest share of the total temperature drop.

Q13 **Answer: 110**

Concept: With one face insulated, dT/dx = 0 there, so the wall behaves like half of a symmetric wall of thickness 2L; the maximum is at the insulated face.

$T_{max} - T_s = q_g L^2/(2k)$ with L = full thickness = 0.05 m

$= 10^6 \times 0.0025/50 = 50$ °C, so $T_{max} =$ **110 °C**

Exam tip: Insulated face = plane of symmetry. Use the full thickness here, the half-thickness for a wall cooled on both sides.

Q14 **Answer: (B) increasing the contact pressure and filling the gaps with a thermal grease**

Concept: Real surfaces touch only at peaks; the gaps (usually air, low k) create the contact resistance $R_c = 1/h_c$.

Higher pressure flattens the peaks (more true contact area); a grease or soft metal foil replaces air in the gaps.

A vacuum or plastic film would increase the resistance.

Exam tip: This is why CPU heat sinks use thermal paste.

Q15 **Answer: (B) largest at the smaller end**

Concept: In steady state with an insulated side, $Q = -kA(x)\, dT/dx$ is the same at every section.

So $|dT/dx| = Q/(kA)$: where A is smallest, the gradient is steepest.

The temperature profile is therefore curved, not linear, even with constant k.

Exam tip: 'Linear profile' needs constant k AND constant area AND no generation.

What next?

Day 41 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 42/90: Week 7 Revision. Comment **GATE90** on our reel to get all links.
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